

Lab Drift Analysis Project

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Problem Statement

The Challenge of Detecting Data Drift in Clinical Labs

The Issue

- Laboratory analyzers measure critical values (hemoglobin, WBC, platelets)
- Calibration drifts gradually, systematic shift from true patient results
- Drift accumulates silently, affects thousands before detection

The Challenge

- How distinguish real patient-mix changes from instrument degradation?
- Standard QC catches obvious failures, not slow drift (0.1–0.5 units/day)
- Drift escapes notice until clinical consequences emerge

Clinical Impact

- 0.3 g/dL hemoglobin bias could withhold needed transfusions or trigger unnecessary interventions with infection/volume risks
- Detection speed and sensitivity are critical for patient safety



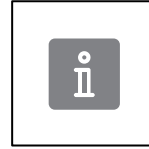
Project Objectives

Goals of Drift Detection Analysis



BACKGROUND

- Analyzer drift affects quality and patient safety across all tests
- Many labs rely on manual QC flagging, reactive approach
- Misses subtle, gradual trends undetectable by eye



IMPORTANCE

- Compare Shewhart vs CUSUM vs EWMA control charts
- Catch drift early
- Enable evidence-based, automated quality monitoring

Methodology Overview

Approach to Detecting Lab Value Drift

1. Downloaded MIMIC-III lab data and filtered to hemoglobin, WBC, and platelets
2. Calculated the average lab value per day for each test
3. Used the first 30 days as a "normal" baseline to understand what typical values look like
4. Built three types of control charts to monitor when values start drifting from that baseline
5. Artificially injected a known drift into the data to test how well each chart method catches it
6. Measured how many days it took each method to raise an alarm after drift began
7. Compared which method was fastest and most sensitive at detecting drift



Dataset Overview

MIMIC-III Dataset & LABEVENTS Description

row_id	unique record identifier
subject_id	patient identifier (links across MIMIC tables)
hadm_id	hospital admission ID (can be NULL for outpatient labs)
itemid	lab test code
charttime	timestamp of when the lab was collected/resulted
value	raw result
valuenum	numeric result for quantitative analysis
valueuom	unit of measure (mEq/L, mg/dL, %, ug/mL, etc.)
flag	abnormality label

Source

- MIMIC-III Clinical Database Demo (v1.4) via PhysioNet
- **100 ICU patients**, de-identified, from Beth Israel Deaconess Medical Center (2001–2012)

Lab Tests Captured

- CBC components: hemoglobin, WBC, platelets
- Basic metabolic panel: sodium, potassium, chloride, bicarbonate, glucose, creatinine, BUN
- Mixed units across tests — preprocessing required to isolate each analyte

Relevance to Drift Detection

- Sequential **charttime** enables ordering measurements chronologically per analyte
- **valuenum** is the target signal for SPC methods (Shewhart, CUSUM, EWMA)
- **flag** field provides a ground-truth abnormality label for validation

Data Preprocessing

Preparing Lab Data for Analysis

- Loaded MIMIC-III LABEVENTS CSV and parsed **charttime** as datetime
- Filtered to **3 target analytes** by **itemid**: Hemoglobin (50811), WBC (51301), Platelets (51265)
- Dropped rows with null **valuenum** (non-numeric/qualitative results)
- Extracted date from **charttime** and computed **daily aggregates per test**: mean, std, min, max, and sample count
- Mapped **itemid** to human-readable test names using a lookup dictionary
- Sorted chronologically by test and date for time-series analysis



Baseline Establishment-

Defining Reference Periods & Normal Ranges

- Split data into two periods: **first 30 days = reference**, remainder = monitoring
- Computed per-test baseline statistics from the reference period: **mean, standard deviation, and average daily spread**
- Stored baseline mean and SD per analyte to serve as the fixed anchor for all three control chart methods
- Reference period represents the "in-control" state when the analyzer was assumed to be properly calibrated
- Baseline stats used to set control limits for Shewhart ($\pm 3\sigma$), CUSUM ($k = 0.5\sigma$, $h = 5\sigma$), and EWMA ($\lambda = 0.2$)



Method 1

Shewhart Method for Drift Detection

How It Works

- Rule: Any daily mean outside $\pm 3\sigma$ triggers alarm
- Independent measurements; violations rare (0.3% normally)
- Simple, intuitive; no memory of past values

Strengths & Weaknesses

- ✓ Easy to understand and implement
- ✗ Ignores trends; slow to detect gradual drift
- ✗ Requires large shift to trigger (≥ 3 SD)

Your Results

- Hemoglobin: 0% alarm rate (baseline too wide)
- WBC: 21.1% alarm rate (330 violations)
- Platelet: 48.5% alarm rate (761 violations)

Interpretation

- Hemoglobin: Limited sensitivity due to sparse reference period
- WBC & Platelet: High alarm rates $\rightarrow$ alarm fatigue risk



Method 2

CUSUM Method for Drift Detection

How It Works

Accumulates small shifts day-by-day, alarm when sum exceeds threshold

Parameters: allowance $k = 0.5\sigma$, decision threshold $h = 5\sigma$

Two-sided: detects both upward and downward drift

Memory of past values $\rightarrow$ catches accumulated drift fast

Strengths

- ✓ Detects on day of onset (lag = 0 days)
- ✓ Optimal for fixed-size step changes
- ✓ Smallest shift magnitude at detection (11.89 units)
- ✓ Best for critical tests where missed drift = patient harm

Weaknesses

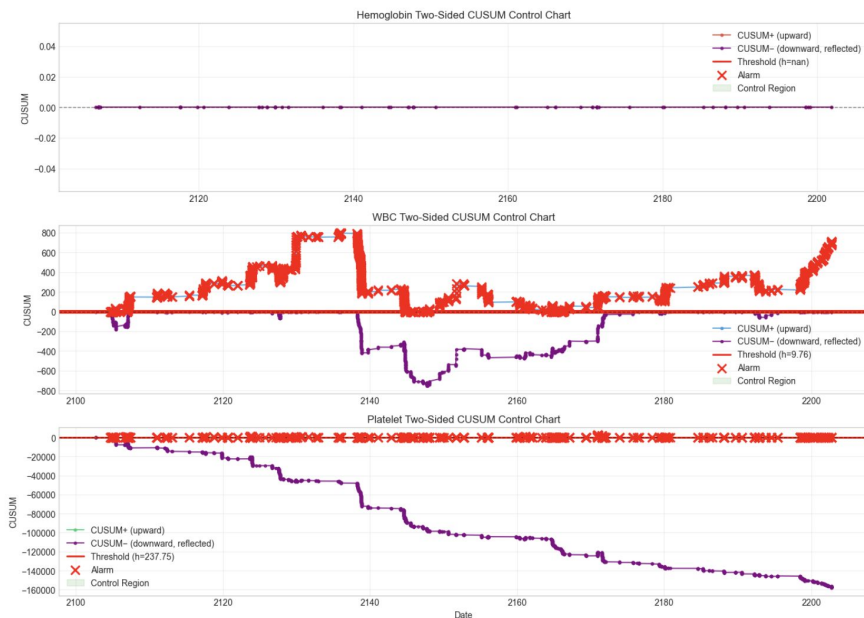
- ✗ Excessive alarms on natural patient variation (99.7%)
- ✗ Can overwhelm lab staff with false positives
- ✗ Requires threshold tuning for each analyte

Your Results

Hemoglobin: 0% alarm rate (no drift to detect)

WBC: 99.7% alarm rate (1,562 violations)

Platelet: 99.7% alarm rate (1,564 violations)



Method 3

EMWA Method for Drift Detection

How It Works

Smoothed average: $\lambda = 0.2$ (20% weight on today, 80% on past)

Reacts to sustained shifts; ignores random noise

Control limits widen over time as algorithm stabilizes

Strengths

- ✓ Balanced: faster than Shewhart, smoother than CUSUM
- ✓ Robust to random noise; reduces false alarms
- ✓ Practical for high-volume analytes in real labs
- ✓ Detection lag = 17 days (acceptable for routine tests)

Weaknesses

- ✗ Slower than CUSUM to detect drift
- ✗ Smoothing delays response to abrupt changes
- ✗ Requires tuning of λ parameter for each analyte

Your Results

Hemoglobin: 0% alarm rate (insufficient baseline data)

WBC: 62.1% alarm rate (973 violations)

Platelet: 83.4% alarm rate (1,307 violations)



Detection Results

Identifying Drift in Lab Values

Ground-Truth Synthetic Drift

Injected upward drift into Platelet counts at day 784

Drift rate: +11.89 K/ μ L per day ($0.25 \times$ baseline SD/day)

Tests which method detects drift earliest and most sensitively

Detection Lag Results

CUSUM: 0-day lag @ 11.89 K/ μ L shift $\rightarrow$ Fastest detection ✓

Shewhart: 2-day lag @ 35.66 K/ μ L shift $\rightarrow$ Requires larger shift

EWMA: 17-day lag @ 213.97 K/ μ L shift $\rightarrow$ Smoothing delays response

Key Findings

CUSUM detects on day of drift onset with minimal magnitude

Shewhart 2-day delay; requires substantial shift

EWMA trades speed for stability; 17 days too slow for critical tests

Comparison of Methods

Performance Comparison across Methods

Metric	Shewhart	CUSUM	EWMA
Simplicity	★★★★★	★★★	★★★★★
Speed	Slow	Fast	Moderate
False Alarms	Low-Med	Very High (99%)	Moderate (62-83%)
Detection Lag	2 days	0 days (Best)	17 days
Best For	Outliers	Critical Tests	Routine Tests
Real-World Use	Supplement	Coagulation, Hgb	WBC, Chemistry

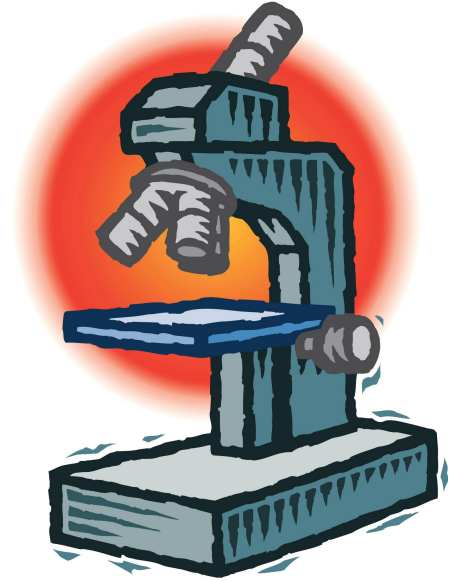
Key Findings

- CUSUM detected drift immediately (0-day lag) and at the smallest shift, making it the most sensitive method
- Shewhart required larger shifts and showed delayed detection, limiting usefulness for gradual drift
- EWMA provided a balance, reducing false alarms but introducing detection delay (~17 days)
- High alarm rates in CUSUM (~99%) suggest risk of alarm fatigue in real-world settings
- No single method is optimal: trade-off exists between speed, sensitivity, and false alarms



Limitations

- Synthetic drift was injected, which may not fully reflect real-world clinical drift patterns
- Limited baseline period (30 days) may reduce accuracy of reference values, especially for hemoglobin
- Analysis focused on only three lab tests, limiting generalizability across all analytes
- Patient variability may influence results, making it difficult to separate true drift from population changes
- Thresholds and parameters were not fully optimized for each analyte





Future Work

- Apply methods to real, unlabeled drift scenarios to validate performance in clinical settings
- Optimize thresholds and parameters for each analyte to reduce false alarms
- Expand analysis to additional lab tests and larger datasets
- Develop hybrid approaches combining methods (e.g., CUSUM + EWMA) for better balance
- Integrate automated alerts into clinical lab systems for real-time monitoring

THANKS

QUESTIONS?



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